Research Article

EDUCATIONAL ATTAINMENT AS A PROTECTIVE FACTOR FOR PSYCHIATRIC DISORDERS: FINDINGS FROM A NATIONALLY REPRESENTATIVE LONGITUDINAL STUDY

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Objective: *This study examined cross-sectional and longitudinal relationships between educational attainment and psychiatric disorders (i.e., mood, anxiety, substance use, and personality disorders) using a nationally representative survey of US adults.* **Method:** *We used data from Waves 1 and 2 of the National Epidemiologic Survey on Alcohol and Related Conditions ($N = 34{,}653$). Bivariate and multiple logistic regressions examined cross-sectional and longitudinal associations between educational attainment and a variety of past-year and incident anxiety, mood, and substance use disorders, controlling for sociodemographics and psychiatric disorder comorbidity.* **Results:** *Adjusted cross-sectional data indicated that educational attainment below a graduate or professional degree at Wave 2 was associated with significantly higher odds of substance use and/or dependence disorders (adjusted odds ratio range (AORR = 1.55–2.55, $P < 0.001$). Longitudinal adjusted regression analyses indicated that individuals reporting less than a college education at Wave 1 were at significantly higher odds of experiencing any incident mood (AORR 1.49–1.64, $P < 0.01$), anxiety (AORR 1.35–1.69, $P < 0.01$), and substance use disorder (AORR 1.50–2.02, $P < 0.01$) at Wave 2 even after controlling for other sociodemographic variables and psychiatric comorbidity.* **Conclusion:** *Findings lend support to other published research demonstrating that educational attainment is protective against developing a spectrum of psychiatric disorders. Mechanisms underlying this relationship are speculative and in need of additional research. Depression and Anxiety 00:1–11, 2016.*

Key words: *education; psychiatric; epidemiology; mood, anxiety; substance use; personality disorder*

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INTRODUCTION

Epidemiological research has frequently examined how psychiatric disorders relate to a variety of social, economic, and cognitive factors such as income, employment, and intellectual functioning.[1–6] Educational attainment has received some attention in relation to risk for psychiatric disorders, with most research suggesting an inverse association with a number of psychiatric disorders such as substance abuse, mood, and anxiety disorders.[3,4,7–9] These findings have emerged primarily from studies consisting of adolescents or young adults and have not been able to consistently take into account other confounding variables such as income, age, or marital status. The lack of more diverse population-representative samples within this body of literature has led to an incomplete understanding of the relationship between educational attainment and psychiatric disorders. Other published literature examining the relationship between intellectual functioning,[10] socioeconomic status, and risk for psychiatric disorders[11,12] may be helpful in contextualizing the relationship between educational attainment and psychiatric disorders.

Evidence suggests that brain structure and function is protective in the development of psychiatric disorders.[10] Published research examining intellectual ability has been relatively consistent in identifying a negative correlation between intellectual ability and the likelihood of having a psychiatric disorder, consistent with the cognitive reserve hypothesis.[10] A 20-year longitudinal cohort study by Koenen et al.[2] indicated: (1) a robust negative association between childhood IQ and risk of schizophrenia, major depression, and anxiety disorders in adulthood; (2) a positive association between childhood IQ and risk for manic episodes; and (3) no relationship between childhood IQ and adult substance abuse/dependence, social phobia, or panic disorder. Other published research has found mixed support for the associations between lower IQ and bipolar disorder[13] substance use/abuse,[14,15] generalized anxiety disorder,[16,17] and PTSD.[18] Mortensen[15] documented a link between personality disorders and low IQ. Given that intelligence is positively correlated with educational attainment,[19,20] one would expect that educational attainment would also be negatively associated with likelihood of having a variety of psychiatric diagnoses. However, it is unclear whether IQ and educational attainment have similar relationships with psychiatric disorders.

The relationship between educational attainment and psychiatric disorders could also be conceptualized by viewing education as a proxy for socioeconomic status. Low socioeconomic status has been thought to create higher risk for stress, adversity, and in turn, psychiatric disorders, from a social causation perspective.[11] Low income and unemployment, which are typically correlated with educational attainment, have also been associated with the prevalence, incidence, and maintenance of common psychiatric disorders.[6,21,22] Lee et al.'s[4] prospective cohort study of disadvantaged, community-dwelling youth indicated that having a high school diploma decreased the odds of having co-occurring psychiatric and substance use problems in young adulthood. Other literature suggests that psychiatric disorders disrupt or hinder attempts to advance education.[11,12] Several published studies indicate that early onset psychiatric problems are associated with premature termination of education, even after controlling for shared familial contributions to educational attainment.[3,7,8,23]

Taken together, published research suggests that lower levels of educational attainment should increase the odds of having certain psychiatric disorders. Nonetheless, relatively few published studies have specifically examined educational attainment in relation to psychiatric disorders with a broader, more diverse sample of adults. In particular, there is a lack of population-representative, longitudinal data that adjust for potential confounding factors on the relationship between educational attainment and psychiatric disorders. These data would clarify to what extent having limited formal education is a unique correlate of specific psychiatric disorders in adulthood. This may assist clinicians in identifying at-risk groups, and perhaps more easily than relying on socioeconomic status and intelligence, which tend to be more difficult to quantify. More importantly, by increasing our understanding of this relationship, we are better situated to target psychiatric disorder prevention programs appropriately and mitigate distress and disability to at risk groups.

OBJECTIVE

The objective of this study is to examine the cross-sectional and longitudinal relationships between educational attainment and psychiatric disorders (i.e., mood, anxiety, substance use disorders, and personality disorders) adjusting for cofounding variables using a nationally representative survey of U.S. adults.

METHOD

SAMPLE

We used survey data from Waves 1 and 2 of the National Epidemiologic Survey on Alcohol and Related Conditions (NESARC), a nationally representative survey of noninstitutionalized adults residing in the United States conducted by the National Institute on Alcohol Abuse and Alcoholism and the National Institutes of Health. Wave 1 was conducted in 2001–02, which included $n = 43{,}093$ participants (response rate of 81%). The sampling frame was based on the 2000 U.S. Census data. Wave 1 included all 50 States and the District of Columbia. The survey also included military personnel living off base and residents in noninstitutionalized group quarters housing, such as boarding houses, shelters, and dormitories.[24] Wave 2 was conducted in 2004–05, which included $n = 34{,}653$ eligible respondents from Wave 1 (response rate of 86.7%). The cumulative response rate of the NESARC was 70.2%, which is the product of Waves 1 and 2 response rates. NESARC weights were employed to ensure sociodemographic representativeness of the U.S. population, as well as to reflect design characteristics of the NESARC and account for nonresponse and nonrandom attrition. Other published research summarizes the demographic characteristics of the NESARC sample[25] and compares it to the general population.[26] Data in both waves were collected using face-to-face, computer-assisted interviews conducted in respondents' homes by trained lay interviewers. Other publications document NESARC's sampling procedures.[27,28]

MEASURES

Psychiatric Diagnoses. The NESARC used the Alcohol Use Disorder and Associated Disabilities Interview Schedule-Diagnostic and Statistical Manual of Mental Disorders-Fourth Edition (AUDADIS-IV) to make DSM-IV diagnoses. The AUDADIS-IV is a semistructured interview that permits trained lay interviewers to yield past-year and lifetime DSM-IV diagnoses. The AUDADIS-IV is a highly reliable and valid instrument for the assessment of psychiatric disorders.[29] Kappa values for mood and anxiety disorders ranges from 0.41 to 0.74 for past-year and 0.42 to 0.70 for lifetime specifiers.[29] The mood disorders included major depressive disorder, dysthymia, mania, and hypomania. In addition to an examination of individual disorders, these diagnoses were combined to create an "any mood disorder" variable. The anxiety disorders included generalized anxiety disorder, posttraumatic stress disorder, panic disorder with and without agoraphobia, agoraphobia without panic disorder, social phobia, and specific phobia. These diagnoses were combined to create an "any anxiety disorder" variable. The drug use disorders included nicotine dependence, alcohol abuse and/or dependence, and an "other" drug abuse and/or dependence category, which included sedatives, tranquilizers, opioids, amphetamines, cannabis, hallucinogens, cocaine, inhalants/solvents, and heroin. Alcohol abuse and/or dependence and the "other drug abuse and/or dependence" categories were combined to create an "any substance abuse and/or dependence disorder (except nicotine)" category. Ten personality disorders were assessed on a lifetime basis in the NESARC at different waves, including antisocial, borderline, narcissistic, histrionic, paranoid, avoidant, dependent, obsessive–compulsive, schizotypal, and schizoid personality disorders. We created a summary variable for "any personality disorder." For the cross-sectional comorbidity analyses, we used past-year psychiatric disorders at Wave 2. For the longitudinal analyses, we derived incident psychiatric disorders by excluding those who met the specified lifetime mental disorder at Wave 1 but subsequently endorsed the criteria for the specified disorder between Waves 1 and 2. Lifetime diagnoses of substance use disorders were not available at Wave 1 so individuals with past-year diagnoses at that time were excluded from these analyses.

Educational Attainment and Other Sociodemographic Variables. We used the sociodemographic variables assessed at Wave 2, with the exception of using Wave 1 educational attainment for the longitudinal analyses. Education was categorized into five groups: less than high school, high school diploma or GED, some college, college or technical school degree, and a graduate or professional degree. The other assessed sociodemographic variables included sex, marital status, age, household income, and ethnicity. Marital status was categorized in three groups: married/cohabitating, divorced/separated/widowed, or never married. Age was categorized into four groups: 20–29, 30–44, 45–64, or 65 and over. Household income was categorized into four groups: \$0–19,999, \$20,000–34,999, \$35,000–69,999, or \$70,000 or more. Race/ethnicity was categorized into five groups: White, Black, American Indian or Alaska Native, Asian/Hawaiian, and Hispanic or Latino. Our categorization of sociodemographic variables was based on previous published literature using the NESARC.[25,30–32]

STATISTICAL ANALYSES

We used the statistical weights accompanying the NESARC dataset and analyzed data using SUDAAN statistical software (version 11.0; Research Triangle Park, NC). SUDAAN permits the use of Taylor Series Linearization, a variance estimation technique suited to large and complex survey sampling methodology.[33] Cases with missing values were deleted list wise. Attrition analyses were conducted to determine whether there were significant differences between Wave 2 respondents and those who only participated in Wave 1 in terms of sociodemographic characteristics, and the prevalence of any lifetime mood, anxiety, or personality disorder, and any past-year substance use disorder at Wave 1. Chi-square analyses were used with response at Wave 2 as the independent variable (present vs. missing) and the aforementioned categorical variables as the dependent variables. Emergent nonrandom differences between Wave 1 and Wave 2 respondents are accounted for through the NESARC sample weights.

We computed cross-tabulations to determine the prevalence of past-year psychiatric disorders at Wave 2 within each category of educational attainment. Cross-tabulations were also calculated to determine the rate of incident (i.e., since last interview) psychiatric disorders at Wave 2 within each education category.

We examined the relationship between educational attainment and past-year psychiatric disorders at Wave 2 using a series of bivariate and multivariable logistic regression models. The first series of logistic regressions was unadjusted (UOR) and included education at Wave 2 as the independent variable and each individual past-year psychiatric disorder at Wave 2 as the dependent variable. The adjusted logistic regression (AOR) model included sociodemographic characteristics and past-year psychiatric disorders diagnosed at Wave 2 as covariates, excluding the disorder category of interest. For example, in the case of social phobia, the adjusted model controlled for sociodemographics, any mood, substance abuse and/or dependence, or personality disorder but excluded any anxiety disorder.

We also examined the relationship between educational attainment and incident psychiatric disorders at Wave 2 using a series of bivariate and multivariable logistic regression models. The first series of logistic regressions was unadjusted and included education categories at Wave 1 as the independent variable and each incident psychiatric disorder (i.e., since Wave 1) as the dependent variable. The multivariable (i.e., AOR) model included sociodemographics and past-year psychiatric disorders at Wave 1 as covariates, excluding the disorder of interest. As indicated, individuals who met criteria for lifetime disorders, or past-year substance use disorders at Wave 1 were also excluded from these analyses.

RESULTS

ATTRITION ANALYSES

Attrition analyses indicated that Wave 2 respondents ($n = 34{,}653$) had a higher prevalence of any lifetime mood ($\chi^2 = 52.50$, $P < 0.001$), anxiety ($\chi^2 = 176.50$, $P < 0.001$), or personality disorder ($\chi^2 = 14.09$, $P < 0.001$) compared to Wave 1 respondents who did not participate in Wave 2 (i.e., those who dropped out between Waves 1 and 2; $n = 8{,}440$). There were no differences observed between Wave 1 and 2 respondents in terms of past-year substance use disorders at Wave 1. Wave 2 respondents also differed significantly from Wave 1 respondents who did not participate in Wave 2 in terms of educational attainment ($\chi^2 = 34.74$, $P < 0.001$),

TABLE 1. Cross-tabulation of educational attainment and past-year psychiatric disorders at Wave 2

	Graduate or professional degree *n* (weighted %)	Bachelor/tech degree *n* (weighted %)	Some college *n* (weighted %)	High school/GED *n* (weighted %)	Less than high school *n* (weighted %)
Any mood disorder	367 (7.3)	778 (9.1)	919 (12.4)	1,053 (10.7)	678 (11.6)
Major depressive disorder	312 (6.4)	633 (7.4)	745 (9.9)	807 (8.3)	526 (8.8)
Dysthymia	33 (0.6)	93 (0.90)	103 (1.2)	129 (1.2)	120 (1.9)
Mania or hypomania	84 (1.7)	233 (2.7)	304 (4.2)	367 (3.7)	242 (4.4)
Any anxiety disorder	586 (12.3)	1,196 (14.4)	1,320 (17.2)	1,525 (14.9)	917 (16.6)
Generalized anxiety disorder	152 (3.3)	297 (3.5)	337 (4.4)	346 (3.4)	231 (4.3)
Posttraumatic stress disorder	156 (3.1)	353 (4.2)	413 (5.1)	467 (4.4)	326 (5.3)
Panic disorder w/ or w/out agoraphobia	72 (1.5)	175 (2.0)	221 (3.1)	225 (2.2)	159 (2.9)
Agoraphobia without panic disorder	6 (0.1)	9 (0.1)	9 (0.1)	9 (0.1)	6 (0.1)
Social phobia	82 (1.7)	191 (2.4)	237 (3.0)	290 (2.7)	145 (2.5)
Specific phobia	288 (6.1)	580 (6.9)	652 (8.6)	757 (7.3)	481 (8.6)
Any substance abuse or dependence (except nicotine)	534 (12.3)	1,446 (19.1)	1,695 (24.1)	2,120 (24.2)	1,110 (22.7)
Nicotine dependence	240 (5.4)	820 (10.9)	1,075 (15.4)	1,513 (17.5)	864 (17.7)
Alcohol abuse and/or dependence	327 (7.6)	780 (10.5)	815 (11.8)	851 (9.5)	369 (7.4)
Other drug abuse and/or dependence	45 (1.1)	153 (2.0)	209 (3.2)	226 (2.6)	115 (2.8)
Personality disorders	858 (17.9)	1,653 (19.5)	1,925 (25.2)	2,214 (21.4)	1,233 (23.1)

Note: Total NESARC Wave 2 sample size $n = 34{,}653$.

income ($\chi^2 = 64.31$, $P < 0.001$), marital status ($\chi^2 = 96.90$, $P < 0.001$), age ($\chi^2 = 99.64$, $P < 0.001$), and ethnicity ($\chi^2 = 8.14$, $P < 0.001$).

CROSS-SECTIONAL ASSOCIATIONS BETWEEN EDUCATIONAL ATTAINMENT AND PSYCHIATRIC DISORDERS

Table 1 displays cross-tabulations of psychiatric disorders by education status at Wave 2 with cell sizes. Overall, we found the highest prevalence rates of psychiatric disorders among adults endorsing some college for any mood disorder (12.4%), any anxiety disorder (17.2%), and any personality disorder (25.2%). In the case of any substance abuse and/or dependence, we found similar prevalence rates for those who endorsed some college (24.1%) and high school/GED (24.2%) as their highest level of educational attainment. Across the assessed psychiatric disorders, graduate or professional degree had the lowest prevalence rates.

Logistic regressions assessing significant associations and effect sizes between educational attainment and past-year psychiatric disorders compared to a graduate or professional degree (reference group) at Wave 2 are presented in Table 2. Bivariate analyses are presented in addition to the multivariable models. After the adjustment of sociodemographic variables and comorbid psychiatric disorders (excluding the disorder of interest), educational attainment below graduate or professional degree was associated with significantly increased odds of any substance abuse and/or dependence excluding nicotine (AORR = 1.55–2.55, $P < 0.001$) and nicotine dependence (AORR = 1.94–4.26, $P < 0.001$). Results further indicated that having a bachelor/technical degree or a high school/GED education was associated with increased odds of an alcohol abuse and/or dependence disorder (AORR = 1.25–1.26, $P < 0.01$) compared to those with a graduate or professional degree. Individuals with a high school/GED or incomplete high school education were also at higher odds for having other drug abuse and/or dependence disorders compared to those

TABLE 2. Logistic regressions examining the association between educational attainment and past-year psychiatric disorders at Wave 2

	Bachelor/tech degree		Some college		High school/GED		Less than high school	
	UOR (99% CI)	AOR (99% CI)	UOR (99% CI)	AOR (99% CI)	UOR (99% CI)	AOR (99% CI)	UOR (99% CI)	AOR (99% CI)
Any mood disorder	1.27	1.01	1.79	1.12	1.53	1.06	1.67	1.03
	(1.02–1.57)*	(0.80–1.28)	(1.50–2.15)**	(0.90–1.40)	(1.27–1.84)**	0.84–1.34)	(1.34–2.08)**	(0.79–1.35)
Major depressive disorder	1.16	0.94	1.62	1.03	1.33	0.93	1.41	0.87
	(0.91–1.48)	(0.73–1.21)	(1.32–1.98)**	(0.82–1.30)	(1.07–1.65)**	(0.72–1.20)	(1.12–1.79)**	(0.65–1.15)
Dysthymia	1.50	1.07	2.07	1.04	2.08	1.08	3.28	1.34
	(0.83–2.71)	(0.59–1.96)	(1.10–3.87)*	(0.54–2.01)	(1.09–3.96)*	(0.56–2.10)	(1.80–5.96)**	(0.68–2.61)
Mania or hypomania	1.63	1.23	2.54	1.42	2.23	1.46	2.66	1.60
	(1.06–2.51)*	(0.79–1.92)	(1.71–3.78)**	(0.92–2.19)	(1.51–3.28)**	(0.94–2.25)	(1.72–4.13)**	(0.98–2.63)
Any anxiety disorder	1.20	1.06	1.49	1.13	1.25	1.01	1.42	1.19
	(1.02–1.42)*	(0.89–1.26)	(1.27–1.74)**	(0.94–1.35)	(1.07–1.47)**	(0.84–1.21)	(1.17–1.74)**	(0.94–1.50)
Generalized anxiety disorder	1.08	0.91	1.37	0.93	1.05	0.75	1.33	0.97
	(0.79–1.47)	(0.66–1.25)	(0.99–1.90)	(0.65–1.35)	(0.79–1.39)	(0.54–1.03)	(0.93–1.91)	(0.62–1.50)
Posttraumatic stress disorder	1.37	1.16	1.67	1.13	1.45	1.00	1.74	1.10
	(1.00–1.89)*	(0.84–1.59)	(1.20–2.32)**	(0.79–1.61)	(1.05–2.01)*	(0.71–1.42)	(1.23–2.45)**	(0.74–1.64)
Panic disorder w/ or w/out	1.34	1.11	2.12	1.44	1.48	1.06	1.96	1.35
agoraphobia	(0.88–2.03)	(0.71–1.73)	(1.36–3.29)**	(0.89–2.33)	(0.99–2.22)	(0.66–1.71)	(1.24–3.11)**	(0.78–2.36)
Agoraphobia without panic	0.98	0.88	0.86	0.61	0.79	0.67	0.59	0.61
disorder	(0.17–5.69)	(0.16–5.04)	(0.16–4.76)	(0.10–3.74)	(0.15–4.23)	(0.10–4.58)	(0.09–3.98)	(0.0.7–5.27)
Social phobia	1.36	1.12	1.77	1.08	1.57	1.10	1.45	0.93
	(0.92–2.01)	(0.74–1.68)	(1.17–2.70)**	(0.68–1.71)	(1.07–2.31)*	(0.72–1.68)	(0.90–2.34)	(0.55–1.58)
Specific phobia	1.14	1.03	1.46	1.17	1.22	1.07	1.46	1.36
	(0.89–1.45)	(0.80–1.32)	(1.16–1.83)**	(0.92–1.50)	(0.95–1.56)	(0.82–1.39)	(1.10–1.93)**	(1.00–1.86)
Any substance abuse and/or	1.69	**1.55**	2.28	**1.91**	2.29	**2.34**	2.10	**2.55**
dependence (except nicotine)	(1.42–2.01)**	**(1.29–1.87)****	(1.89–2.75)**	**(1.55–2.36)****	(1.95–2.68)**	**(1.96–2.80)****	(1.70–2.59)**	**(2.01–3.23)****
Nicotine	2.12	**1.94**	3.19	**2.66**	3.69	**3.52**	3.74	**4.26**
dependence	(1.65–2.73)**	**(1.50–2.51)****	(2.44–4.17)**	**(2.01–3.52)****	(2.92–4.67)**	**(2.73–4.25)****	(2.82–4.98)**	**(3.16–5.74)****
Alcohol abuse and/or	1.42	**1.25**	1.62	1.26	1.28	**1.26**	0.96	1.06
dependence	(1.16–1.73)**	**(1.01–1.55)***	(1.30–2.00)**	(0.98–1.60)	(1.06–1.54)**	**(1.02–1.56)***	(0.77–1.21)	(0.80–1.40)
Other drug abuse and/or	1.88	1.47	3.03	1.71	2.45	**1.92**	2.68	**2.13**
dependence	(1.02–3.44)*	(0.78–2.77)	(1.77–5.19)**	(0.97–3.03)	(1.41–4.26)**	**(1.04–3.52)***	(1.42–5.03)**	**(1.11–4.14)***
Personality disorders	1.11	0.92	1.55	1.11	1.25	0.96	1.38	1.01
	(0.96–1.30)	(0.78–1.08)	(1.31–1.82)**	(0.92–1.33)	(1.08–1.45)**	(0.82–1.14)	(1.14–1.66)**	(0.82–1.25)

Note: UOR, unadjusted odds ratio; AOR, adjusted for sociodemographics and any anxiety, mood, or substance abuse/dependence disorder excluding the independent variable.
Bolded odds ratios denote statistically significant findings in the adjusted models; $^{*}P < 0.01$, $^{**}P < 0.001$; reference group = graduate or professional degree.

with a graduate or professional degree (AORR = 1.92–2.13, $P < 0.01$).

LONGITUDINAL ASSOCIATIONS BETWEEN EDUCATIONAL ATTAINMENT AND PSYCHIATRIC DISORDERS

Table 3 displays cross-tabulations of educational attainment at Wave 1 by incident psychiatric disorders at Wave 2, including cell sizes. Results partially corroborated cross-sectional findings. Graduate or professional degree had the lowest overall prevalence rates across the assessed psychiatric disorders. Further, attaining some college education had the highest prevalence rate of substance abuse and/or dependence excluding nicotine (9.1%).

Findings of the bivariate and multivariate logistic regression analyses examining the association between educational attainment at Wave 1 and incident psychiatric disorders are displayed in Table 4. In the adjusted model, some college, high school/GED, and less than high school were significantly associated with increased odds of any incident mood disorder (AORR = 1.49–1.64, $P < 0.01$), any incident anxiety disorder (AORR = 1.35–1.69, $P < 0.01$), any incident substance abuse, and/or dependence excluding nicotine (AORR = 1.65–2.02, $P < 0.01$), and nicotine dependence (AORR = 2.92–5.41, $P < 0.001$) compared to graduate or professional degree. Results were nonsignificant in the case of the association between bachelor/technical degree and incident any mood disorder and any anxiety disorder.

DISCUSSION

This study is unique in its examination of the cross-sectional and longitudinal relationships between educational attainment and psychiatric disorders, adjusting for a number of cofounding variables, using a nationally representative survey of U.S. adults. The cross-sectional findings indicated that lower levels of education were associated with increased odds of having a substance

TABLE 3. Cross-tabulation of educational attainment at Wave 1 and incident psychiatric disorders

	Graduate or professional degree *n* (weighted %)	Bachelor/tech degree *n* (weighted %)	Some college *n* (weighted %)	High school/GED *n* (weighted %)	Less than high school *n* (weighted %)
Any mood disorder	166 (4.7)	417 (6.3)	435 (8.2)	668 (8.2)	404 (8.4)
Major depressive disorder	152 (4.2)	373 (5.3)	372 (6.5)	568 (6.8)	342 (6.7)
Dysthymia	22 (0.4)	78 (0.9)	71 (0.9)	90 (0.9)	92 (1.5)
Mania or hypomania	53 (1.3)	185 (2.4)	231 (3.6)	330 (3.4)	215 (3.9)
Any anxiety disorder	238 (6.5)	505 (8.2)	529 (9.5)	707 (8.6)	416 (9.4)
Generalized anxiety disorder	128 (3.0)	256 (3.4)	286 (4.1)	325 (3.3)	215 (4.1)
Posttraumatic stress disorder	41 (0.9)	91 (1.3)	93 (1.2)	129 (1.3)	88 (1.4)
Panic disorder w/ or w/out agoraphobia	66 (1.6)	143 (2.0)	193 (2.7)	246 (2.5)	166 (3.1)
Agoraphobia without panic disorder	6 (0.1)	11 (0.2)	8 (0.1)	13 (0.1)	7 (0.1)
Social phobia	54 (1.2)	140 (1.8)	173 (2.5)	234 (2.3)	122 (2.0)
Specific phobia	182 (4.6)	433 (6.0)	442 (6.3)	615 (6.4)	403 (7.6)
Any substance abuse or dependence (except nicotine)	100 (3.8)	294 (6.2)	386 (9.1)	424 (6.2)	218 (5.4)
Nicotine dependence	81 (2.2)	320 (5.1)	382 (7.0)	694 (9.3)	384 (8.9)
Alcohol abuse and/or dependence	94 (3.5)	276 (5.6)	371 (8.2)	416 (6.0)	209 (4.9)
Other drug abuse and/or dependence	25 (0.7)	99 (1.5)	140 (2.6)	143 (1.7)	85 (1.9)

Note: Total NESARC wave 2 sample size $n = 34{,}653$.

abuse and/or dependence disorder of varying types (i.e., nicotine, alcohol, other). Longitudinal analyses supported the robust nature of these associations and demonstrated a broader relationship between educational attainment and other psychiatric disorders. Specifically, educational attainment at Wave 1 was significantly associated with a variety of incident psychiatric disorders at Wave 2 even after controlling for other sociodemongraphic indicators and comorbid psychiatric disorders. Specifically, adults with less than a college education were at increased odds of incident any mood disorder, panic disorder, social and other phobias, and any substance abuse and/or dependence disorder, including nicotine dependence. Readers are advised that when comparing findings across different conditions, odds ratios should not be interpreted as relative risks but rather as heightened probabilities in relation to those with a graduate or professional degree. That being said, these findings suggest that having higher education is a unique protective factor for a wide range of psychiatric disorders. Nonetheless, the correlational nature of this data prohibits causal conclusions, therefore we will speculate on several potential mechanisms that could underlie this relationship that have also been alluded to in previous published research.

First, consistent with the cognitive reserve hypothesis, IQ and educational attainment may share common neurological and/or genetic etiological factors with psychiatric disorders.[2] For example, Koenen et al. speculated that low childhood IQ and by implication, educational attainment, could be related to neuroanatomical differences that also increase risk for psychiatric disorders.[2] IQ has been found to be significantly correlated with intracranial, cerebral, temporal lobe, hippocampal, and cerebellar volume.[33,34] Likewise, a large body of published research has documented structural abnormalities and neurophysiological correlates of psychiatric disorders such as bipolar disorder, major depressive disorder, and borderline personality disorder.[36–39] Unfortunately, we were unable to assess IQ as a confounding variable within the present study. As such, additional research identifying the shared neurological correlates of educational attainment and psychiatric disorders is needed.

Second, individuals with lower levels of education could be exposed to other risk factors for psychiatric

TABLE 4. Logistic regressions examining the association between educational attainment at Wave 1 and incident psychiatric disorders

	Bachelor/tech degree		Some college		High school/GED		Less than high school	
	UOR (99% CI)	AOR (99% CI)	UOR (99% CI)	AOR (99% CI)	UOR (99% CI)	AOR (99% CI)	UOR (99% CI)	AOR (99% CI)
Any mood disorder	1.37 (1.01–1.87)*	1.24 (0.90–1.71)	1.81 (1.37–2.40)**	**1.49** **(1.10–2.02)***	1.81 (1.37–2.39)**	**1.63** **(1.20–2.22)****	1.86 (1.36–2.55)**	**1.64** **(1.17–2.31)****
Major depressive disorder	1.27 (0.92–1.76)	1.13 (0.80–1.59)	1.57 (1.16–2.12)**	1.29 (0.92–1.79)	1.65 (1.22–2.23)**	**1.46** **(1.04–2.06)***	1.63 (1.17–2.28)**	1.39 (0.96–2.01)
Dysthymia	2.54 (1.14–5.64)*	2.18 (0.96–4.94)	2.63 (1.23–5.60)*	1.84 (0.80–4.24)	2.49 (1.12–5.52)*	1.87 (0.78–4.47)	4.22 (2.06–8.64)**	**2.66** **(1.17–6.05)***
Mania or hypomania	1.94 (1.16–3.25)*	1.71 (0.99–2.95)	2.88 (1.75–4.74)**	**2.08** **(1.20–3.63)****	2.73 (1.67–4.45)**	**2.27** **(1.28–4.02)****	3.13 (1.88–5.22)**	**2.59** **(1.43–4.68)****
Any anxiety disorder	1.28 (0.96–1.70)	1.23 (0.91–1.66)	1.50 (1.16–1.94)**	**1.37** **(1.02–1.83)***	1.35 (1.03–1.76)*	**1.35** **(1.00–1.83)***	1.48 (1.10–1.99)**	**1.69** **(1.20–2.38)****
Generalized anxiety disorder	1.15 (0.84–1.58)	1.07 (0.78–1.46)	1.40 (0.98–2.02)	1.16 (0.80–1.69)	1.11 (0.83–1.50)	1.00 (0.72–1.37)	1.38 (0.96–1.97)	1.23 (0.81–1.86)
Posttraumatic stress disorder	1.33 (0.69–2.59)	1.15 (0.58–2.29)	1.26 (0.65–2.42)	0.94 (0.47–1.89)	1.34 (0.71–2.52)	1.07 (0.54–2.10)	1.44 (0.77–2.69)	1.09 (0.54–2.19)
Panic disorder w/ or w/out agoraphobia	1.30 (0.73–2.29)	1.22 (0.68–2.20)	1.71 (1.10–2.68)*	1.46 (0.89–2.38)	1.59 (1.02–2.48)*	1.51 (0.92–2.49)	2.00 (1.27–3.14)**	2.13 (1.27–3.58)**
Agoraphobia without panic disorder	1.59 (0.33–7.59)	1.76 (0.39–7.98)	0.83 (0.16–4.39)	0.82 (0.14–4.77)	1.13 (0.26–4.94)	1.34 (0.28–6.39)	1.08 (0.18–6.53	1.60 (0.20–12.62)
Social phobia	1.61 (0.99–2.64)	1.45 (0.88–2.37)	2.20 (1.36–3.54)**	**1.67** **(1.00–2.80)***	2.02 (1.26–3.23)**	**1.73** **(1.03–2.89)***	1.77 (1.04–3.00)*	1.55 (0.87–2.76)
Specific phobia	1.33 (1.01–1.76)**	1.27 (0.95–1.58)	1.41 (1.09–1.87)**	1.25 (0.94–1.67)	1.42 (1.09–1.87)**	**1.36** **(1.01–1.82)***	1.71 (1.26–2.32)**	**1.76** **(1.24–2.49)****
Any substance abuse and/or dependence (except nicotine)	1.68 (1.19–2.37)**	**1.50** **(1.05–2.13)***	2.54 (1.78–3.62)**	**2.02** **(1.40–2.92)****	1.70 (1.20–2.40)**	**1.65** **(1.16–2.34)**??	1.45 (0.96–2.19)	**1.74** **(1.12–2.69)***
Nicotine dependence	2.37 (1.56–3.58)**	**2.20** **(1.45–3.35)****	3.32 (2.19–5.03)**	**2.92** **(1.91–4.46)****	4.54 (3.07–6.70)**	**4.44** **(3.00–6.68)****	4.32 (2.92–6.41)**	**5.41** **(3.59–8.14)****
Alcohol abuse and/or dependence	1.65 (1.16–2.34)**	**1.45** **(1.02–2.08)***	2.49 (1.72–3.59)**	**1.94** **(1.33–2.84)****	1.77 (1.24–2.53)**	**1.67** **(1.16–2.40)****	1.43 (0.95–2.17)	**1.63** **(1.04–2.53)***
Other drug abuse and/or dependence	2.24 (1.10–4.53)*	1.82 (0.90–3.68)	3.87 (1.96–7.66)**	**2.44** **(1.21–4.91)***	2.47 (1.27–4.80)**	1.94 (0.99–3.83)	2.84 (1.37–5.86)**	**2.57** **(1.21–5.45)***

Note: UOR, unadjusted odds ratio; AOR, adjusted for sociodemographics and any anxiety, mood, or substance abuse/dependence disorder excluding the independent variable.
Bolded odds ratios denote statistically significant findings in the adjusted models; $^*P < 0.01$, $^{**}P < 0.001$; reference group = graduate or professional degree.

disorders such as stress.[4] Lower levels of education may make it more difficult to obtain employment and secure income, which may increase stress and propensity to develop psychiatric disorders. Conversely, individuals with higher levels of education have been shown to have a greater degree of occupational advancement and income stability,[40,41] and could conceivably have more access to social support.[42,43] The combination of economic and social resources may help buffer the effects of stress and protect against certain psychiatric disorders.

Third, individuals with lower levels of education may also have less access to quality mental health care and health insurance. An estimated 11% of Americans do not have health insurance, which includes 26% from the lowest income quintile.[44] Low income individuals in the United States are more likely to have unmet health needs.[44] By extension, individuals with low education may not have access to appropriate and timely psychiatric services. Relatedly, low levels of educational attainment have been associated with poor mental health literacy,[45,46] which includes components such as knowledge of how to prevent/reduce risk of psychiatric disorders, recognition of symptomatology that may be indicative of a developing disorder, knowledge of help-seeking options and treatments available, and knowledge of effective self-help strategies for milder problems.[47] Mental health literacy has implications for when and what type of treatment is sought, symptom management, and prevention of psychiatric problems.[47]

Alternatively, certain personality characteristics may predispose individuals to both seek higher education and be less likely to develop psychiatric disorders. Longitudinal data point to the positive influence of conscientiousness and the adverse role of neuroticism in academic achievement performance.[48,49] Similarly, different personality profiles have been documented between individuals with diagnosed psychiatric disorders and those without.[50] Further research in the area of personality characteristics, educational attainment, and risk for psychiatric disorders would be helpful in clarifying this relationship, especially in light of the findings here that personality pathology was unrelated to educational attainment examined within this study after adjusting for comorbid psychiatric disorders.

Taken together, our findings could be explained by shared neurological and/or genetic etiological factors, higher levels of stress that can accompany socioeconomic disadvantage, limited mental health literacy and/or mental health care, and personality characteristics. It is also possible that other factors that were not reviewed here could be influencing our findings.

Our findings are an important contribution to the literature given that little published research has specifically examined educational attainment in relation to psychiatric disorders. Most recent research has examined sociodemographic status as an aggregate construct or specific aspects of sociodemographics. For example, Hudson[5] demonstrated an indirect impact of sociodemographic status on mental health via economic hardship in middle and low income groups. We were able to demonstrate education's unique impact on psychiatric disorders even after controlling for income. Furthermore, whereas previous research has used restricted age ranges[4,51,52] this study provides an indication of the relationship between educational attainment and psychiatric disorders using a diverse age range, allowing us to concurrently control for any confounding influence of age.

Collectively, our results have broad implications for psychiatric disorder prevention programs. Despite not being able to determine causality in our findings, we highlight the importance of delivering psychiatric disorder prevention programs to individuals with lower levels of education. Existing mental health promotion initiatives and prevention/early intervention programs have focused preferentially on socioeconomically disadvantaged populations.[53–55] Increasing efforts to target mental health promotion and early intervention to individuals with limited educational attainment are encouraged. An examination of the health literacy literature more broadly provides some direction of where mental health promotion efforts could be focused.[56] Having limited formal education is associated with greater difficulties with communicative and critical health literacy, meaning individuals may lack skill in extending new health information to changing situations as well as appraising information critically to exert control over health situations.[56] Assisting individuals to improve both aspects of health literacy, particularly as it applies to mental health, could be beneficial to those with limited education.

Interventions could also focus on educational attainment as a protective agent for incident psychiatric disorders, in line with other literature citing that education is one of the strongest predictors of physical health.[57–60] Prevention efforts within this area could include increasing efforts to reduce dropout rates and creating more opportunities for educational or vocational training among those with psychiatric disorders. For example, programs designed to improve access to education and prevent premature dropouts from high school show efficacy in reducing adolescent mental health difficulties,[23,61,62] also supporting the social causation explanation.

LIMITATIONS AND FUTURE RESEARCH

These findings should be considered in light of several limitations. This study was unable to examine other relevant variables that could potentially influence the relationship between psychiatric disorders and educational attainment, such as psychiatric and educational familial history, self-reported stress, IQ, other psychosocial factors and personality characteristics. The study was also limited in its ability to examine other psychiatric disorders (e.g., psychotic disorders) in relation to education. NESARC data also relies on retrospective recall of age of onset of psychiatric disorders, which poses a threat to the reliability our findings. Additionally, although the reliability of the AUDADIS-IV for variety of different disorders has been demonstrably good to excellent[27,63] the mania/hypomania diagnosis has been studied only to a limited extent.[29] However, the existing literature suggests that the dimensional scale reliability for manic disorder is good.[29] Relatedly, given the time frame of NESARC data collection, only DSM-IV diagnoses are available. Future research would be well advised to explore how mediating variables, such as family background, personality characteristics and self-reported stress, might influence the relationship between educational attainment and DSM-5 psychiatric disorders. Given that the NESARC excluded institutionalized adults, conducting similar analyses in institutionalized samples (e.g., incarcerated populations, psychiatric hospitals, nursing homes) would provide interesting supplements to whether our findings would replicate. Given the large volume of analyses conducted within our study, a corresponding risk of type I error is elevated. We adopted a more stringent P value of 0.01 and used 99% confidence intervals to help reduce, albeit not eliminate, the probability of false-positive results. Finally, respondent attrition between Wave 1 and Wave 2 may introduce nonrandom bias into our results given that Wave 2 nonresponders differed significantly in terms of the prevalence of several categories of psychiatric disorders and sociodemographic variables. However, NESARC sample weights are designed to help correct for nonrandom respondent attrition and ensure that Wave 2 respondents are still well representative of the US population.[26]

CONCLUSION

This study adds to a small body of research examining the relationship between educational attainment and psychiatric disorders using both cross-sectional and longitudinal methods in a nationally representative sample of US adults. Our findings are consistent with other published research documenting that higher education is protective for a wide range of psychiatric disorders after controlling for other aspects of SES, sociodemographics, and psychiatric comorbidity. Future research exploring the mechanisms underlying this robust relationship is recommended. Targeted psychiatric disorder prevention programs to individuals with lower levels of education may be helpful in reducing risk for incident mental health problems as may programs that promote access to educational opportunities.

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